Predictive Modelling

2026-03-26

Importing Libraries

```
library(tidyverse)
library(caret)
library(glmnet)
library(forecast)
```

Importing and Cleaning Data

```
df <- read.csv("Economy_of_US.csv")
df <- df[, c(1, 2, 3, 6, 7, 8, 9)]
colnames(df)[2] <- "GDP"
colnames(df)[3] <- "GDP_PerCapita"
```

Transforming and Grouping Training and Testing Data

```
df <- df %>%
  mutate(
    Year_Scaled = Year - min(Year),      # normalize year
    Year_Sq = Year_Scaled^2,             # polynomial term
    Lag1 = lag(GDP, 1),          # lag feature
    Growth = (GDP / Lag1 - 1) * 100
  ) %>%
  drop_na()

set.seed(123)

train_index <- createDataPartition(df$GDP, p = 0.8, list = FALSE)
train <- df[train_index, ]
test <- df[-train_index, ]
```

Linear, Polynomial, Ridge and Lasso Modelling

```r
lm_model <- lm(GDP ~ Year_Scaled + Lag1 + Growth, data = train) #Initial Linear Regression Model

summary(lm_model)
```

```
##
## Call:
## lm(formula = GDP ~ Year_Scaled + Lag1 + Growth, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -474.91  -33.96   -4.99   46.25  398.40
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -853.13153  111.99291  -7.618 6.21e-09 ***
## Year_Scaled   27.35601   10.03659   2.726  0.00995 **
## Lag1           1.00349    0.01768  56.773  < 2e-16 ***
## Growth       142.62672   13.79385  10.340 3.51e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 160.4 on 35 degrees of freedom
## Multiple R-squared:  0.9996, Adjusted R-squared:  0.9996
## F-statistic: 2.939e+04 on 3 and 35 DF,  p-value: < 2.2e-16
```

```r
# Predictions
lm_pred <- predict(lm_model, newdata = test)


poly_model <- lm(GDP ~ Year_Scaled + Year_Sq, data = train)
poly_pred <- predict(poly_model, newdata = test)

# Prepare matrix format
x_train <- model.matrix(GDP ~ Year_Scaled + Year_Sq + Lag1 + Growth, train)[,-1]
y_train <- train$GDP

x_test <- model.matrix(GDP ~ Year_Scaled + Year_Sq + Lag1 + Growth, test)[,-1]
y_test <- test$GDP

# Ridge (alpha = 0)
ridge_model <- cv.glmnet(x_train, y_train, alpha = 0)
ridge_pred <- predict(ridge_model, s = "lambda.min", newx = x_test)

# Lasso (alpha = 1)
lasso_model <- cv.glmnet(x_train, y_train, alpha = 1)
lasso_pred <- predict(lasso_model, s = "lambda.min", newx = x_test)


rmse <- function(actual, pred) {
  sqrt(mean((actual - pred)^2))
}
```

```r
results <- data.frame(
  Model = c("Linear", "Polynomial", "Ridge", "Lasso"),
  RMSE = c(
    rmse(y_test, lm_pred),
    rmse(y_test, poly_pred),
    rmse(y_test, ridge_pred),
    rmse(y_test, lasso_pred)
  )
)

print(results)
```

```
##        Model     RMSE
## 1     Linear 238.7048
## 2 Polynomial 516.7115
## 3      Ridge 433.0670
## 4      Lasso 421.1631
```

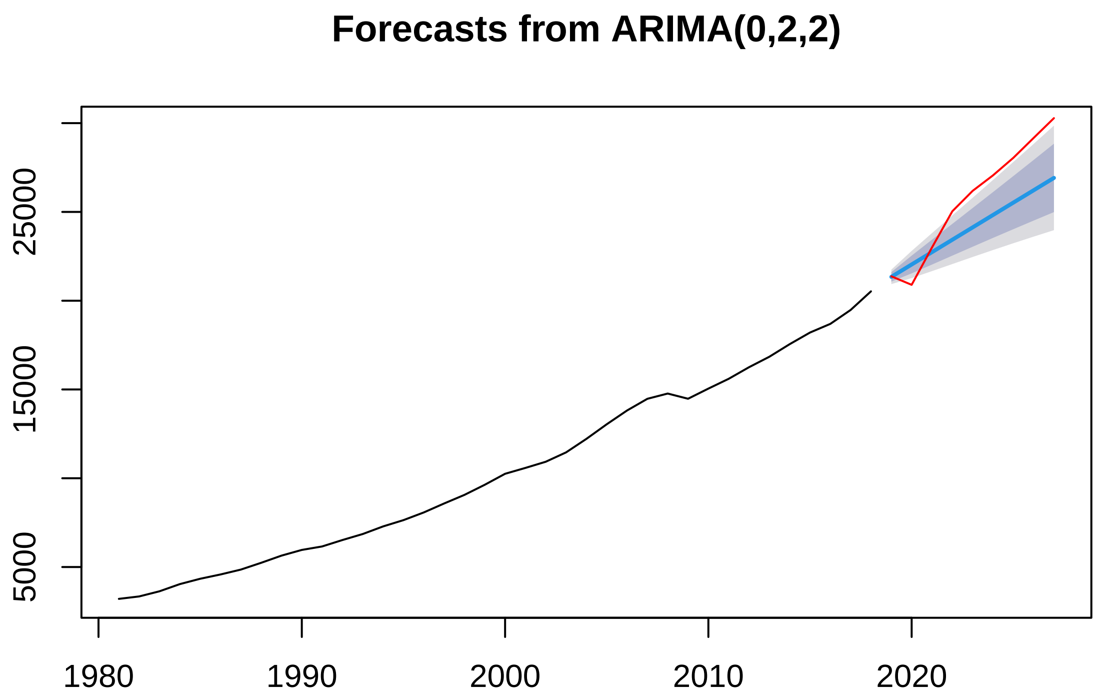
Time Series Forecasting

```r
gdp_ts <- ts(df$GDP, start = min(df$Year), frequency = 1)

train_ts <- window(gdp_ts, end = 2018) #Splitting Data into Testing and Training Data Groups
test_ts  <- window(gdp_ts, start = 2019)

arima_model <- auto.arima(train_ts)

summary(arima_model)
```

```
## Series: train_ts
## ARIMA(0,2,2)
##
## Coefficients:
##           ma1      ma2
##       -0.4080  -0.3417
## s.e.   0.1504   0.1358
##
## sigma^2 = 43099:  log likelihood = -242.5
## AIC=491   AICc=491.75   BIC=495.75
##
## Training set error measures:
##                    ME     RMSE      MAE       MPE     MAPE      MASE
## Training set 50.43472 196.3734 135.4919 0.5534932 1.281003 0.2800077
##                     ACF1
## Training set -0.02022467
```

```r
forecast_values <- forecast(arima_model, h = length(test_ts))

# Plot forecast vs actual
```

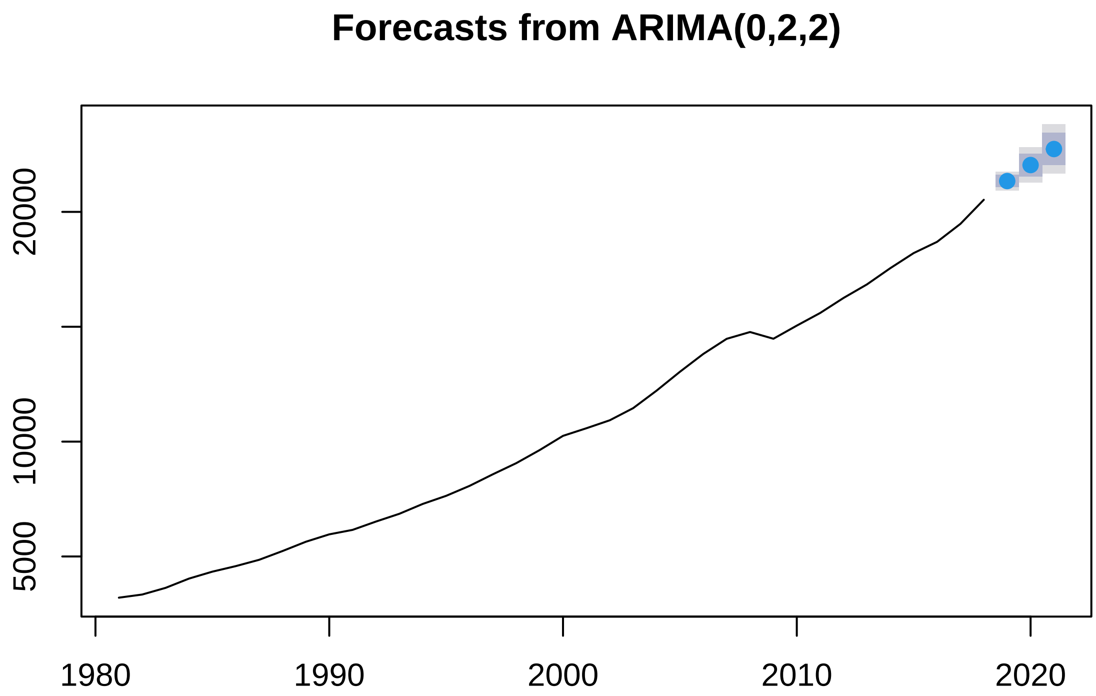
```r
plot(forecast_values)
lines(test_ts, col = "red")
```




```r
future_forecast <- forecast(arima_model, h = 3)

print(future_forecast)
```

```
##      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 2019       21341.67 21075.62 21607.72 20934.77 21748.57
## 2020       22038.44 21538.25 22538.63 21273.47 22803.41
## 2021       22735.21 22034.91 23435.51 21664.19 23806.23
```

```r
plot(future_forecast)
```




```
future_years <- data.frame(
  Year = 2028:2030
)

future_years <- future_years %>%
  mutate(
    Year_Scaled = Year - min(df$Year),
    Year_Sq = Year_Scaled^2,
    Lag1 = tail(df$GDP, 1),
    Growth = mean(df$Growth, na.rm = TRUE)
  )

lm_future_pred <- predict(lm_model, newdata = future_years)

print(lm_future_pred)
```

```
##        1        2        3
## 31559.01 31586.37 31613.72
```

```
# Displaying each Predicted Values from each model

comparison <- data.frame(
  Year = 2028:2030,
  ARIMA = future_forecast$mean,
  Linear_Model = lm_future_pred
```

```
)

print(comparison)
```

```
##   Year    ARIMA Linear_Model
## 1 2028 21341.67     31559.01
## 2 2029 22038.44     31586.37
## 3 2030 22735.21     31613.72
```